

Vital signs: incidence and trends of infection with pathogens transmitted commonly through food--foodborne diseases active surveillance network, 10...



BACKGROUND: In the United States, contaminated food causes approximately 1,000 reported disease outbreaks and an estimated 48 million illnesses, 128,000 **METHODS:** The Foodborne Diseases Active Surveillance Network (FoodNet) conducts surveillance among 15% of the U.S. population for laboratory-confirmed infections with nine pathogens transmitted commonly through food. Overall and pathogen-specific changes in incidence were estimated from 1996-1998 to 2010 and from 2006-2008 to 2010.hospitalizations, and 3,000 deaths annually. This report summarizes 2010 surveillance data and describes trends since 1996. **RESULTS:** A total of 19,089 infections, 4,247 hospitalizations, and 68 deaths were reported from FoodNet sites in 2010. Salmonella infection was the most common infection reported (17.6 illnesses per 100,000 persons) and was associated with the largest number of hospitalizations (2,290) and deaths (29); no significant change in incidence of Salmonella infection has occurred since the start of surveillance during 1996-1998. Shiga toxin-producing Escherichia coli (STEC) O157 infection caused 0.9 illnesses per 100,000. Compared with 1996-1998, overall incidence of infection with six key pathogens in 2010 was 23% lower, and pathogen-specific incidence was lower for Campylobacter, Listeria, STEC O157, Shigella, and Yersinia infection but higher for Vibrio infection. Compared with a more recent period, 2006--2008, incidence in 2010 was lower for STEC O157 and Shigella infection but higher for Vibrio infection. **CONCLUSIONS:** The incidence of STEC O157 infection has declined to reach the 2010 national health objective target of ≥ 1 case per 100,000. This success, as well as marked declines since 1996-1998 in overall incidence of six key foodborne infections, demonstrates the feasibility of preventing foodborne illnesses. **IMPLICATIONS FOR PUBLIC HEALTH PRACTICE:** Salmonella infection should be targeted because it has not declined significantly in more than a decade, and other data indicate that it is one of the most common foodborne infections, resulting in an estimated \$365 million in direct medical costs annually. The prevention measures that reduced STEC O157 infection need to be applied more broadly to reduce Salmonella and other infections. Effective measures from farm to table include preventing contamination of meat during slaughter and of all foods, including produce, during processing and preparation; cooking meat thoroughly; vigorously detecting and investigating outbreaks; and recalling contaminated food.